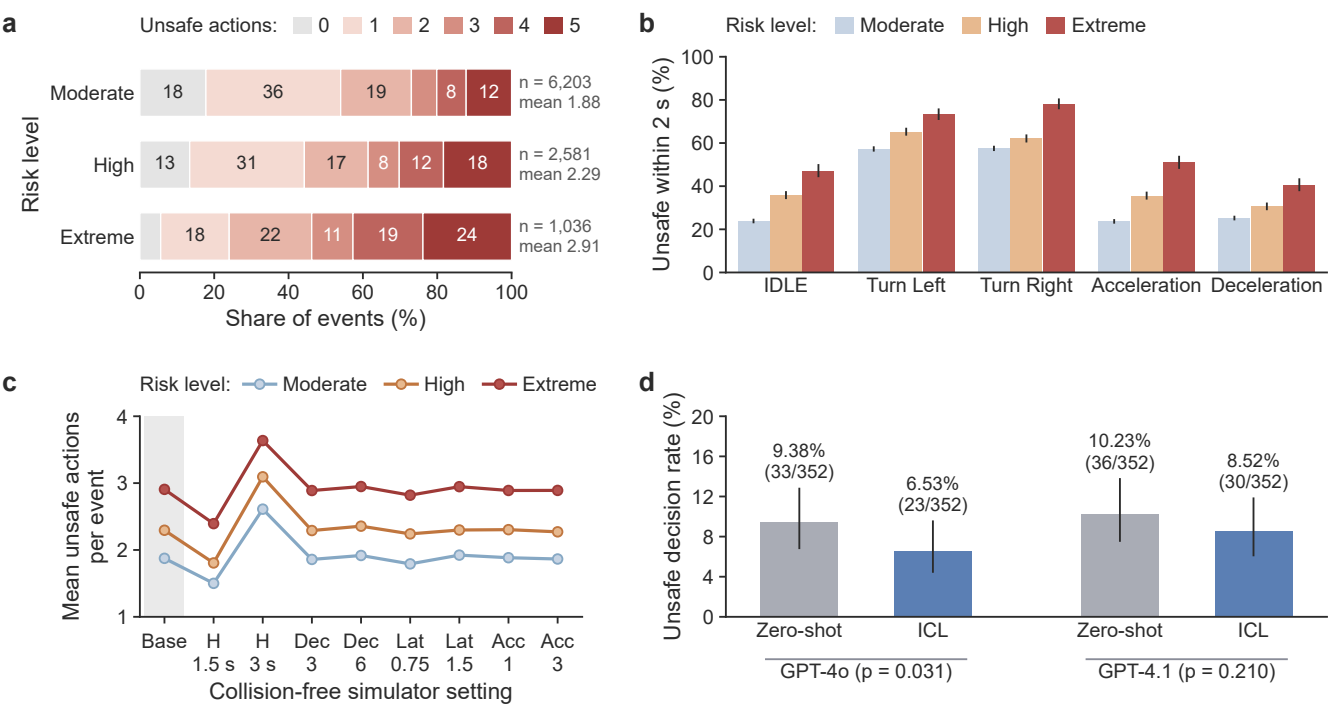
a
Unsafe actions: 0 1 2 3 4 5
Risk level
Moderate
High
Extreme
18
36
19
8
12
n = 6,203
mean 1.88
13
31
17
8
12
18
n = 2,581
mean 2.29
18
22
11
19
24
n = 1,036
mean 2.91
0
20
40
60
80
100
Share of events (%)
b
Risk level: Moderate High Extreme
Unsafe within 2 s (%)
100
80
60
40
20
0
IDLE
Turn Left
Turn Right
Acceleration
Deceleration
c
Risk level: Moderate High Extreme
Mean unsafe actions per event
4
3
2
1
Base
H 1.5 s
H 3 s
Dec 3
Dec 6
Lat 0.75
Lat 1.5
Acc 1
Acc 3
Collision-free simulator setting
d
Unsafe decision rate (%)
20
16
12
8
4
0
9.38% (33/352)
6.53% (23/352)
10.23% (36/352)
8.52% (30/352)
Zero-shot
ICL
Zero-shot
ICL
GPT-4o (p = 0.031)
GPT-4.1 (p = 0.210)